

KATLEGO BARAYI

Software Engineer

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience in full stack development, enterprise platform engineering, API design, observability, and AI/ML prototyping. Proven ability to design, develop, test, deploy, and maintain scalable software solutions at MTN Group, one of Africa's largest telecommunications companies. Proficient across multiple languages (TypeScript, JavaScript, Python, Java, C#) and frameworks (Vue 3, React, Node.js, Express, Spring Boot, FastAPI). Experienced in RESTful API design, enterprise authentication (OAuth 2.0, Okta), CI/CD automation (Docker, GitLab CI, GitHub Actions), observability (Datadog APM, structured logging), and comprehensive testing (115+ tests). Additional experience building AI powered accessibility solutions using MediaPipe, TensorFlow, and NLP APIs at Lelapa AI. Certified SAFe Scrum Master with strong cross functional collaboration skills.

PROFESSIONAL EXPERIENCE

Software Engineer

January 2025 to March 2026

MTN Centre of Excellence, Johannesburg

Contributed across three teams within MTN's Centre of Excellence: CMS Core (platform engineering), Hive (HR content management), and MTN One (CMS production support), serving content operations across MTN's pan African footprint.

CMS Core Team

- Designed and developed a scalable CMS platform using Directus and TypeScript, enabling flexible content modeling and delivery across consumer facing applications for 10+ internal stakeholders.
- Built 5+ custom Directus extensions integrating internal and external REST APIs, streamlining complex business workflows and reducing manual content operations by an estimated 30%.
- Architected and built a Datadog observability extension for Directus, implementing agentless APM tracing, structured logging via Winston, secrets management backed by a database table, HTTP request middleware, and change tracking audit hooks for data mutations.
- Refactored a monolithic Vue 3 video file interface component into a modular architecture (types, composables, utils), improving maintainability and enabling independent testing of each module.
- Developed a comprehensive Vitest test suite with 115+ unit and integration tests, significantly improving code coverage and reducing regressions across the CMS extension ecosystem.
- Implemented enterprise authentication and authorization flows using OAuth 2.0 and Okta, ensuring compliance with organisational security standards across all CMS access points.
- Maintained and improved CI/CD pipelines (GitHub Actions, GitLab CI, Docker) supporting reliable automated deployments, reducing release cycle friction and minimising downtime.
- Instrumented PostHog analytics across multiple frontend modules, enabling data driven decisions on content editor workflows and platform usability improvements.

Hive Team (HR)

- Built and delivered an HR focused CMS powering the Hive platform, enabling management of employee records, quick links, company news, upcoming events, theme of the week, Viva Engage content, articles, training resources, Human Capital Management (HCM), and IT service information.

MTN One Team (CMS Support)

- Provided Tier 1 CMS production support, troubleshooting incidents and enabling content editors, improving system usability and stakeholder satisfaction.
- Applied Agile/Scrum principles as a Certified SAFe Scrum Master, facilitating sprint planning, daily stand ups, retrospectives, and backlog refinement.

Volunteer Coding Mentor

September 2024 to June 2025

MTN Group, Johannesburg

- Volunteered to lead coding workshops with high school students, teaching programming fundamentals and sharing career insights from WeThinkCode_ to professional software engineering at MTN.
- Mentored learners on problem solving approaches and introduced them to pathways in technology, encouraging participation in STEM fields.

Prototype Developer

June 2024 to October 2024

Lelapa AI, Johannesburg

Joined post hackathon to build a production ready Sign Language Interpreter prototype, translating South African Sign Language (SASL) into all 11 of South Africa's official languages.

- Designed and developed an AI powered Sign Language Interpreter in Python, integrating Lelapa AI's NLP language APIs to deliver real time text and audio output, improving accessibility for deaf and hard of hearing users.
- Integrated multilingual translation pipelines supporting dynamic user selected output across all 11 official South African languages, expanding the solution's real world reach.
- Built intuitive user facing React interfaces enabling seamless language and format selection, resulting in a polished prototype recognised as a hackathon finalist.
- Collaborated cross functionally with AI engineers, UX designers, and product stakeholders to rapidly iterate features and validate technical concepts under tight deadlines.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Java, C#, Dart, HTML5, CSS3, SQL

Frontend: React, Vue 3, Flutter, responsive design, component driven UI development

Backend & APIs: Node.js, Express, Javalin, Spring Boot, FastAPI, RESTful API design, custom Directus extensions, API security

CMS & Auth: Directus, OAuth 2.0, Okta, enterprise authentication and authorization

DevOps & CI/CD: Docker, GitHub Actions, GitLab CI, Git, automated deployment pipelines

Observability: Datadog (APM tracing, telemetry, structured logging, instrumentation), PostHog analytics

Testing: Vitest, unit testing, integration testing, UAT, test automation

AI & ML: TensorFlow, PyTorch, scikit learn, MediaPipe (hand landmark extraction), FastAPI, NLP API integration

Tools: Git, GitHub, GitLab, Postman, VS Code, Winston

Methodologies: Agile, Scrum (Certified SAFe Scrum Master), SAFe, sprint planning, backlog refinement

EDUCATION

System Development Diploma

WeThinkCode_, Johannesburg

CERTIFICATIONS

- Certified SAFe Scrum Master
- WeThinkCode_ GenAI Course for Software Engineers

AWARDS & RECOGNITION

- Winner | Nerdma Systems Hack AI thon 2024: AI for Loss Prevention (1st place)
- Finalist | Lelapa AI Hackathon 2024: South African Sign Language Interpreter